JENKINS CI/CD PROJECT

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Course: DevOps

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JENKINS CI/CD PROJECT IMPLEMENTATION

1. Project Overview

This project demonstrates a complete CI/CD (Continuous Integration and Continuous Deployment) pipeline using Jenkins integrated with multiple DevOps tools.

The pipeline automates the following processes:

- Code checkout from GitHub
- Build process using Maven
- Running unit tests
- Code quality analysis using SonarQube
- Deployment to Tomcat server

JENKINS CI/CD PROJECT IMPLEMENTATION

2. Tools Used

- Jenkins
- Apache Maven
- SonarQube
- Apache Tomcat

Plugins Used

- Deploy to Container
- Pipeline Stage View
- SonarQube Scanner
- SonarQube Quality Gates

JENKINS CI/CD PROJECT IMPLEMENTATION

Architecture / Workflow

The workflow of the project is as follows:

1. Developer pushes code to GitHub repository
2. Jenkins pulls the code automatically
3. Maven builds the application
4. Unit testing is performed
5. SonarQube analyzes code quality
6. Application is deployed to Tomcat server

JENKINS CI/CD PROJECT IMPLEMENTATION

4. Jenkins Installation

The following steps were performed to install Jenkins:

•Installed Git and Maven
•Added Jenkins repository
•Installed Java 17
•Installed Jenkins
•Started Jenkins service

Output: Jenkins server started successfully

JENKINS CI/CD PROJECT IMPLEMENTATION

SonarQube Setup

The following steps were performed:

- Downloaded SonarQube package
- Extracted the files
- Installed Java
- Created sonar user
- Started SonarQube service

Output: SonarQube dashboard accessible

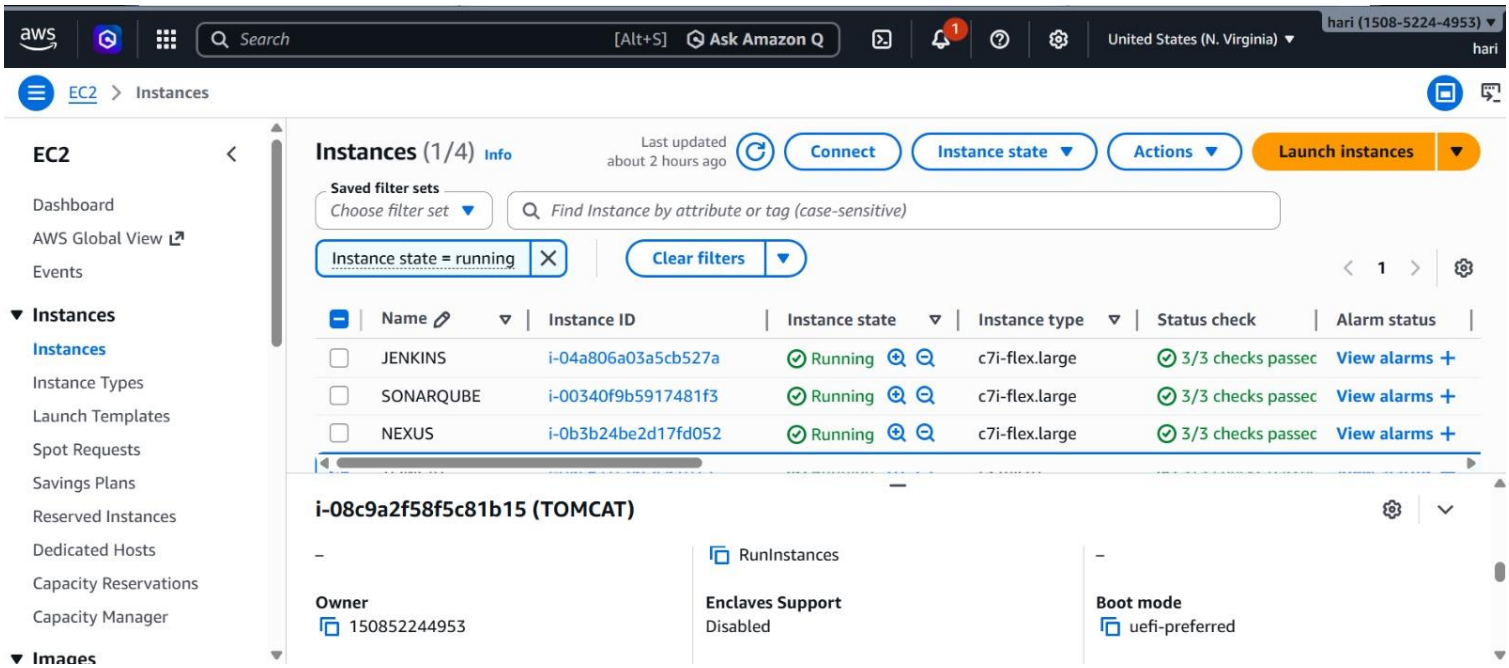
JENKINS CI/CD PROJECT IMPLEMENTATION

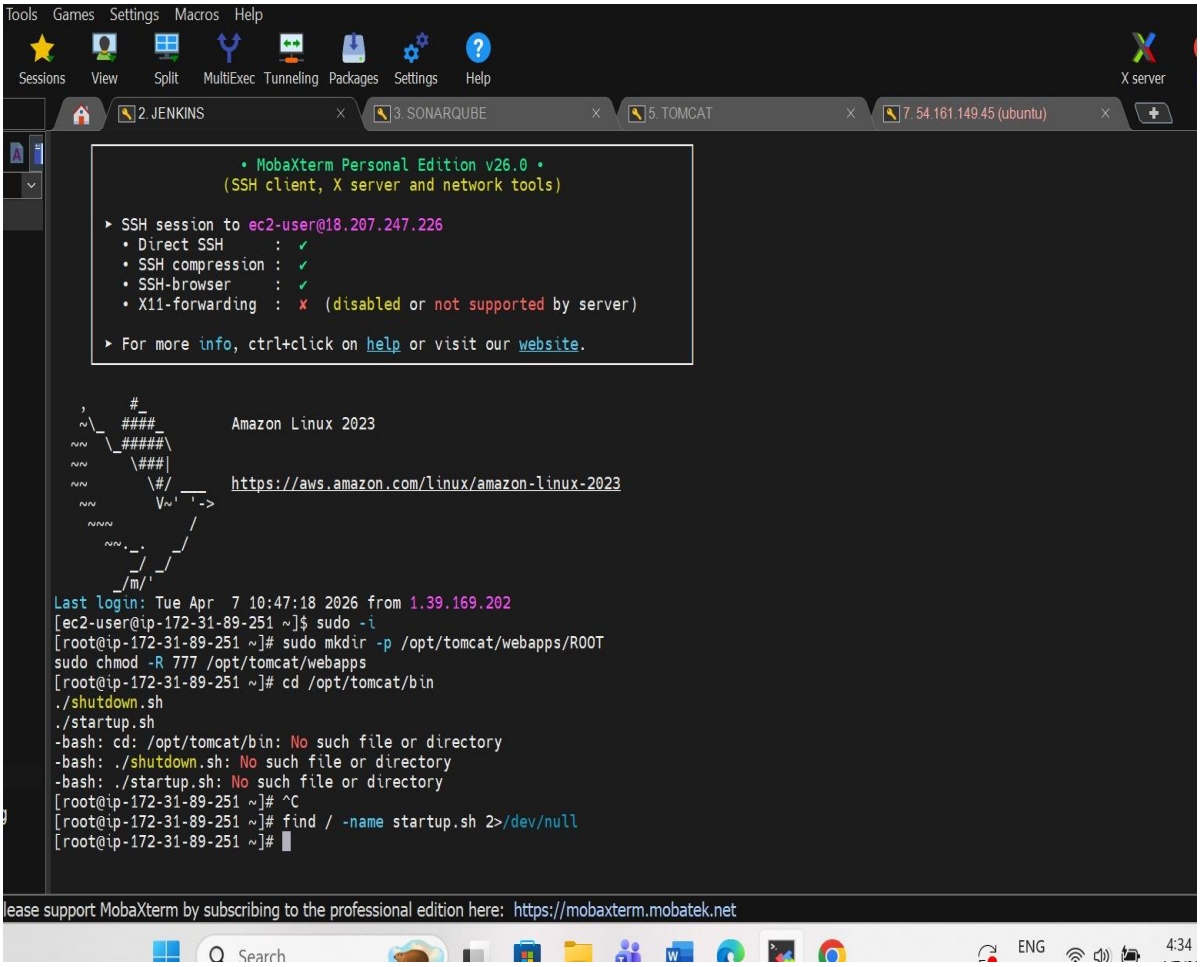
Execution Screenshots

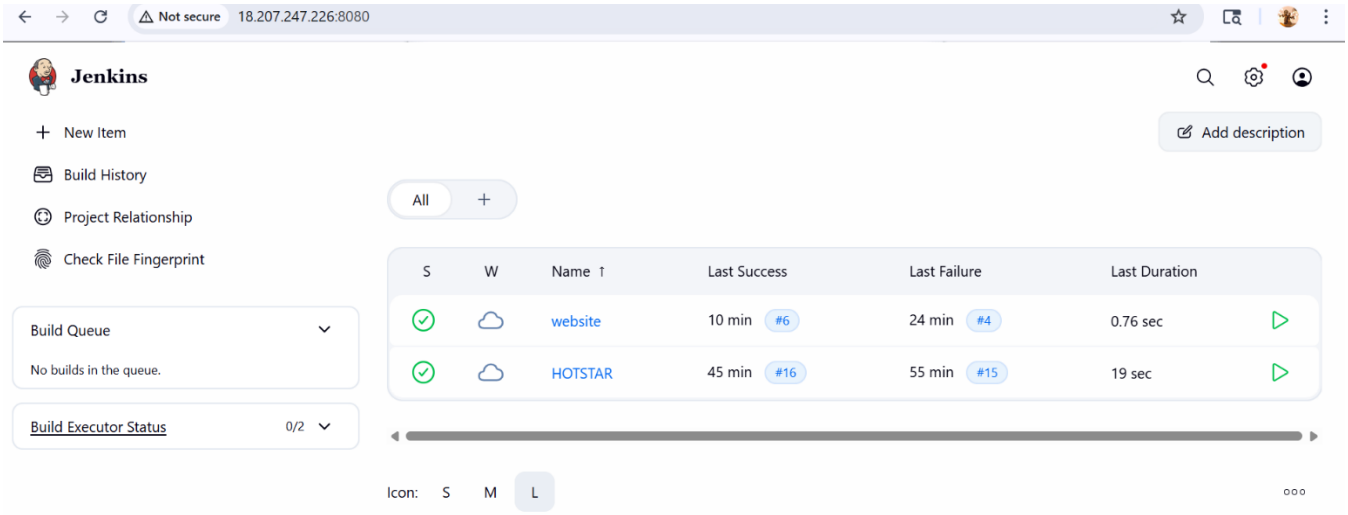
- Jenkins Dashboard
- Pipeline Build Success
- Console Output
- SonarQube Results
- Deployment Output

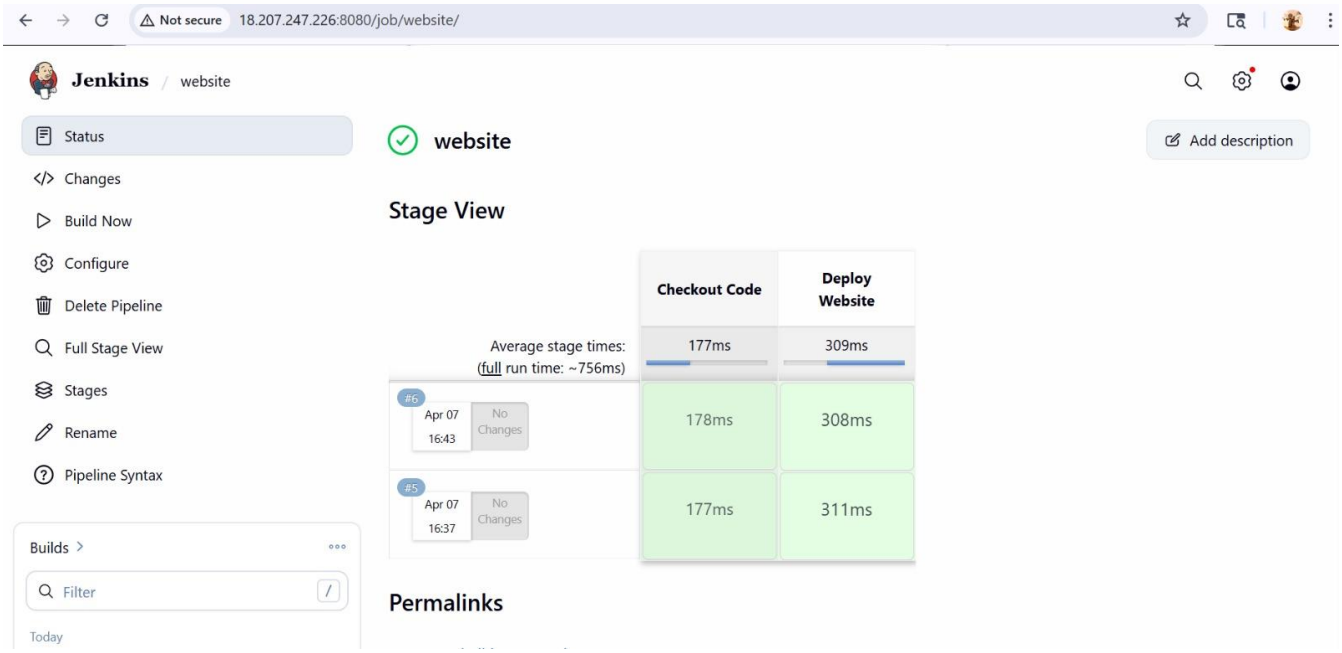
Important Note

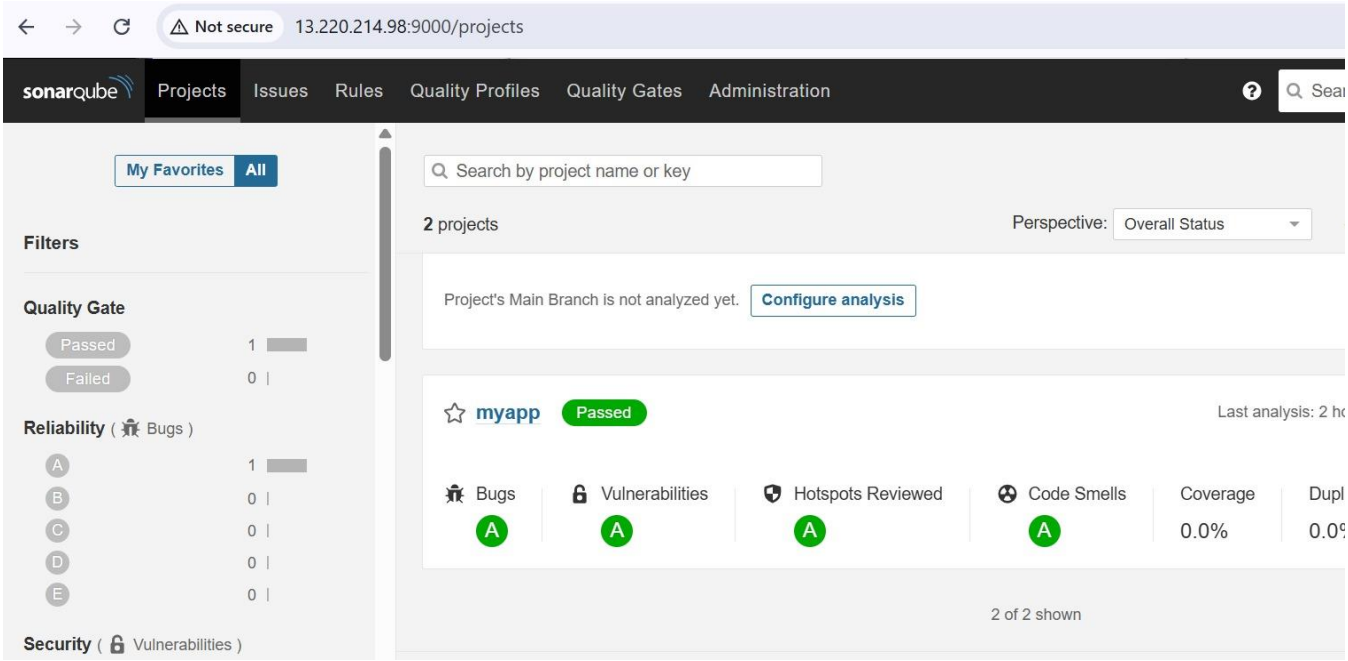
The project was successfully executed on AWS EC2.
Currently, the instance is terminated, so only screenshots are provided as proof of implementation.


aws
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
hari (1508-5224-4953)
hari
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
Instances (1/4) Info
Last updated about 2 hours ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
JENKINS i-04a806a03a5cb527a Running c7i-flex.large 3/3 checks passed View alarms
SONARQUBE i-00340f9b5917481f3 Running c7i-flex.large 3/3 checks passed View alarms
NEXUS i-0b3b24be2d17fd052 Running c7i-flex.large 3/3 checks passed View alarms
i-08c9a2f58f5c81b15 (TOMCAT)
RunInstances
Owner
150852244953
Enclaves Support
Disabled
Boot mode
uefi-preferred


Tools Games Settings Macros Help
Sessions View Split MultiExec Tunneling Packages Settings Help X server
2. JENKINS
3. SONARQUBE
5. TOMCAT
7. 54.161.149.45 (ubuntu)
• MobaXterm Personal Edition v26.0 •
(SSH client, X server and network tools)
▸ SSH session to ec2-user@18.207.247.226
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✗ (disabled or not supported by server)
▸ For more info, ctrl+click on help or visit our website.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Tue Apr 7 10:47:18 2026 from 1.39.169.202
[ec2-user@ip-172-31-89-251 ~]$ sudo -i
[root@ip-172-31-89-251 ~]# sudo mkdir -p /opt/tomcat/webapps/ROOT
sudo chmod -R 777 /opt/tomcat/webapps
[root@ip-172-31-89-251 ~]# cd /opt/tomcat/bin
./shutdown.sh
./startup.sh
-bash: cd: /opt/tomcat/bin: No such file or directory
-bash: ./shutdown.sh: No such file or directory
-bash: ./startup.sh: No such file or directory
[root@ip-172-31-89-251 ~]# ^C
[root@ip-172-31-89-251 ~]# find / -name startup.sh 2>/dev/null
[root@ip-172-31-89-251 ~]#
lease support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
Search
ENG
4:34


Not secure 18.207.247.226:8080
Jenkins
New Item
Build History
Project Relationship
Check File Fingerprint
Add description
All
S W Name ↑ Last Success Last Failure Last Duration
website 10 min #6 24 min #4 0.76 sec
HOTSTAR 45 min #16 55 min #15 19 sec
Build Queue
No builds in the queue.
Build Executor Status 0/2
Icon: S M L


Not secure 18.207.247.226:8080/job/website/
Jenkins / website
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
website
Add description
Stage View
Checkout Code
Deploy Website
Average stage times:
(full run time: ~756ms)
177ms
309ms
#6 Apr 07 16:43 No Changes
178ms
308ms
#5 Apr 07 16:37 No Changes
177ms
311ms
Permalinks

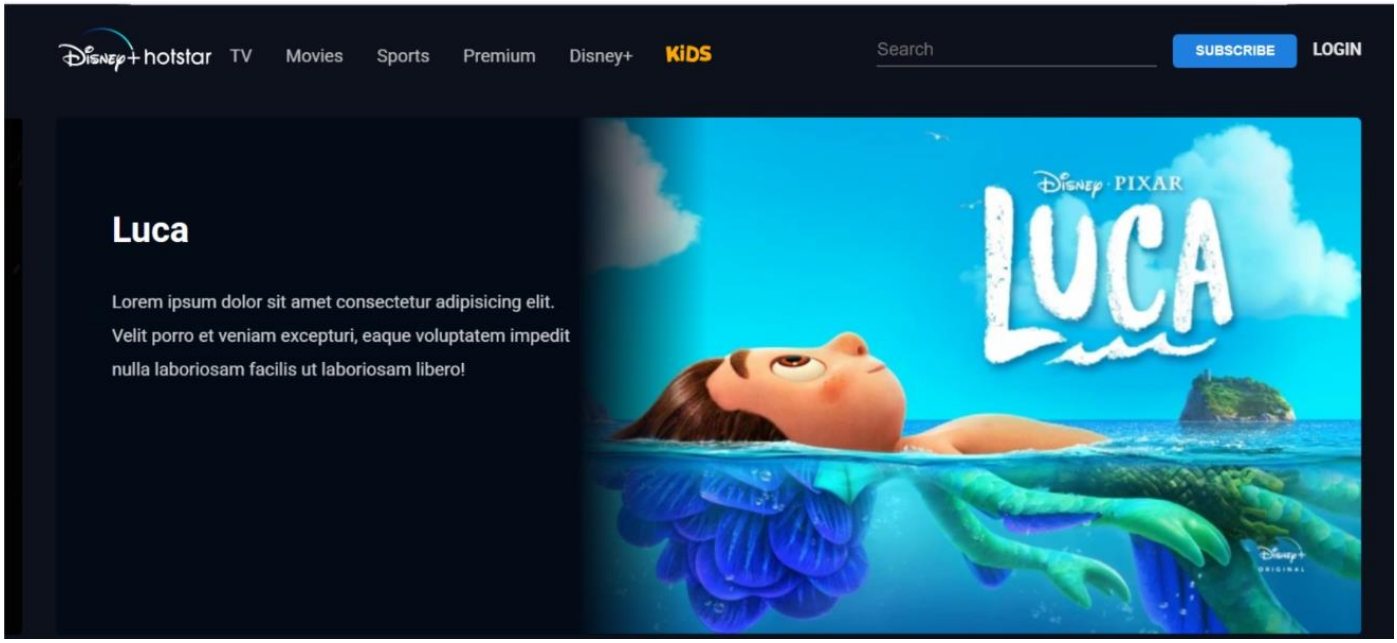

Not secure 13.220.214.98:9000/projects
sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
My Favorites
All
Filters
Quality Gate
Passed 1
Failed 0
Reliability (Bugs)
A 1
B 0
C 0
D 0
E 0
Security (Vulnerabilities)
Search by project name or key
2 projects
Perspective: Overall Status
Project's Main Branch is not analyzed yet.
Configure analysis
myapp
Passed
Last analysis: 2 h
Bugs
Vulnerabilities
Hotspots Reviewed
Code Smells
Coverage
0.0%
2 of 2 shown

Not secure 3.236.239.238:8080/manager/html

Tomcat Web Application Manager

Message:	OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

34°C ENG 4:56 PM